DO NOW — RECALL

# Operating systems and assembly challenge

Name: _____ Class: _____ Date: _____

**1. Which OS component queues print requests?**

**2. What does LMC INP do to ACC?**

**3. What does Waiting mean?**

**4. Rising-edge D flip-flop has Q=1 and D=0 immediately before the active edge. What is Q after?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

A print spooler.

### 2. Model answer

Replaces ACC with supplied input.

### 3. Model answer

Blocked until an event such as I/O completes.

### 4. Model answer

Q=0; it samples D at the clock edge.

### Vocabulary — know these terms

ACC — Accumulator

Ready — Eligible for CPU

Running — Using CPU

Waiting — Blocked on event

NEW LEARNING

# Read, annotate and explain

## Understand

LMC INP overwrites the accumulator; STA saves the value to a mailbox.

## Trace

Processes change between Ready, Running and Waiting as the OS schedules and responds to I/O.

## Explain

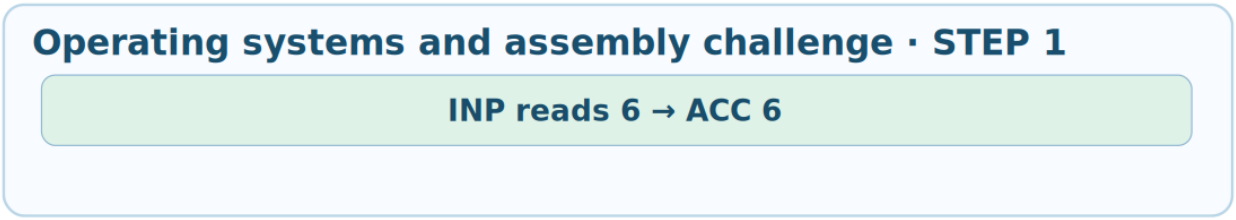
Spooling enables multiple programs to submit print jobs without monopolising the printer.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

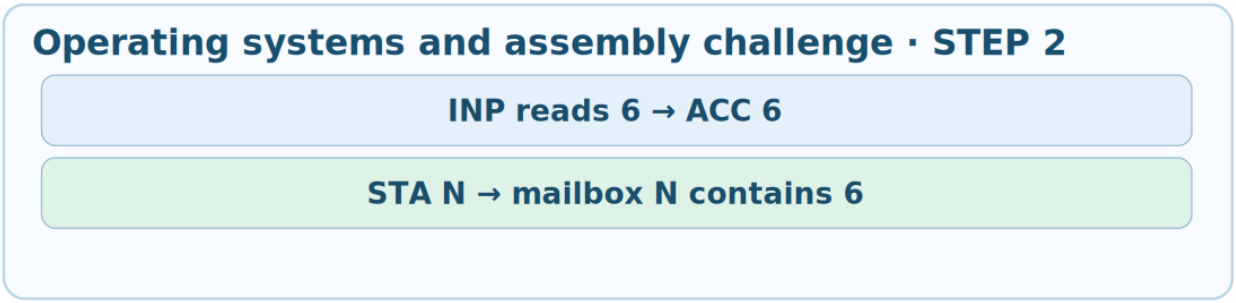
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**



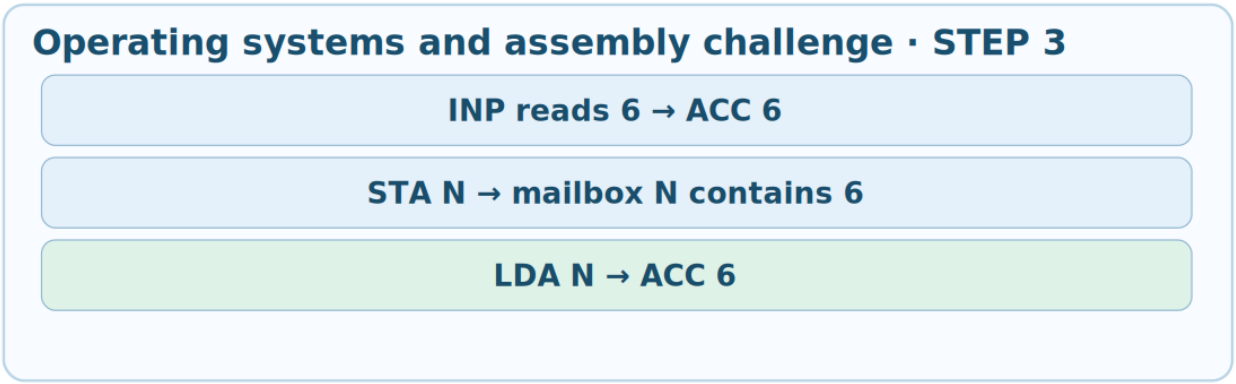

Operating systems and assembly challenge · STEP 1

**New Learning**




Operating systems and assembly challenge · STEP 2

**New Learning**

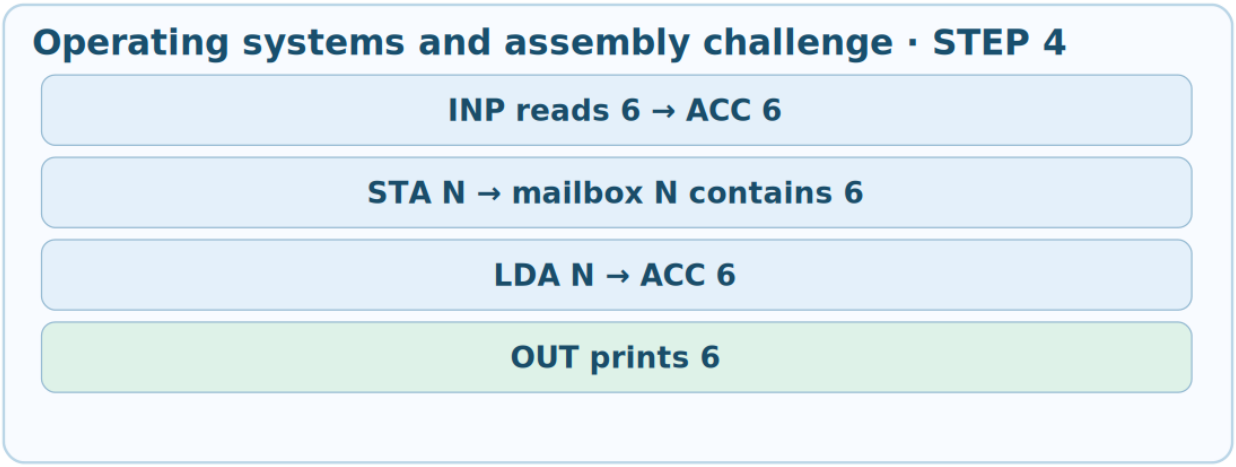



Operating systems and assembly challenge · STEP 3

# New Learning · original lesson diagrams

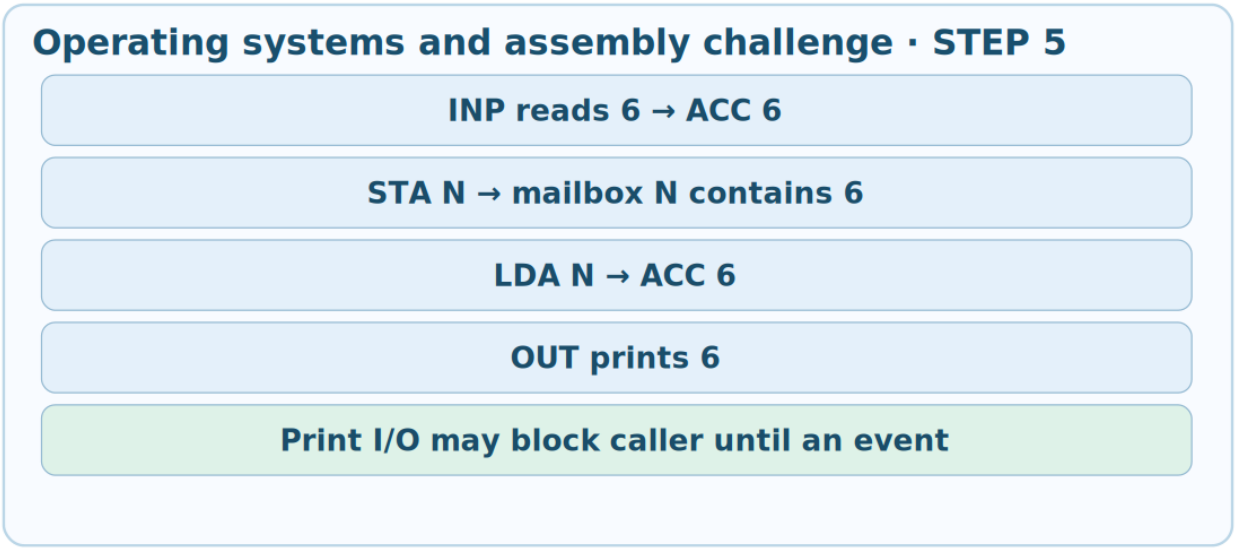
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Operating systems and assembly challenge · STEP 4

**New Learning**




Operating systems and assembly challenge · STEP 5

I DO — WORKED EXAMPLE

# Follow the worked method

### Trace an LMC fragment with input 8.

### Predict first

INP sets ACC=8; STA X writes 8 to mailbox X; the ACC remains 8. HLT ends the program.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Trace an LMC fragment with input 8.

INP sets ACC=8; STA X writes 8 to mailbox X; the ACC remains 8. HLT ends the program.




IDO · STATE TRACE

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

A Running process requests a disk read. Give its next states.
Source code / input:
Running → Waiting → Ready → Running

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

State ACC and X after input 8 followed by INP; STA X. [4]

### Question 2

Describe Ready, Running, Waiting and return to Ready during a disk read. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

Describe OS file access, I/O process state and spooling when an app prints saved data. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**A blocked process continues using the CPU until its I/O completes.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

## GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

## Guided 1 — CHECK

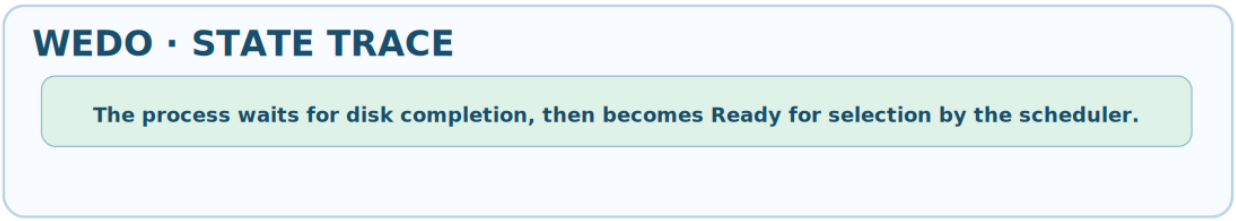
Predict first
The process waits for disk completion, then becomes Ready for selection by the scheduler.

# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A Running process requests a disk read. Give its next states.

The process waits for disk completion, then becomes Ready for selection by the scheduler.




WEDO · STATE TRACE